

ACKNOWLEDGEMENT

The sense of contentment and satisfaction that accompanies the successful completion of this project would be incomplete without mentioning the people who have helped us throughout this journey. Their constant guidance, support, and encouragement made this project possible.

We take this opportunity to thank **Dr.S.B.Biradar**, Principal of S.V.M Arts, Science & Commerce College, **ILKAL**, for providing a healthy academic environment that helped us focus on our work.

We express our deep sense of gratitude to **Shri.P.D.Ekbote**, Head of the Department of Physics, for providing the inspiration and support required for the successful completion of this project.

We would also like to thank our guide **Miss.B.M.Nadaf** and co-guide **Shri.H. M.Meti** for valuable guidance and continuous support throughout the project.

Last but not least, we sincerely thank our friends who helped us in the successful completion of this project and made it more enjoyable.

DECLARATION

We hereby declare that this dissertation entitled "TRANSISTOR SOLAR TRACKER". Based on the results of the experiment and observation carried out by us the guidance of **Miss B.M. Nadaf** and co-guidance of **Shri. H.M. METI**, Department of physics, SVM Arts, Science and Commerce college **ILKAL**. This project is submitted in partial fulfilment of the academic requirements.

We further declare that this work is original and has not been submitted previously, either in part or full, to any other institution of board for the award of any degree.

All information and data included in this project have been collected from the reliable source and proper acknowledgement has been given wherever is necessary.

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ABSTRACT

As energy demand and environmental concerns continue to rise, natural energy sources have become increasingly important as alternatives to conventional energy sources. The renewable energy sector is rapidly growing as a key area of development for many countries due to its environmental and economic benefits. Among these, solar energy plays a vital role as a primary energy source, especially in rural areas.

This project aims to develop a system that tracks the movement of the sun to achieve maximum energy efficiency using a 555 IC and LDR (Light Dependent Resistor) sensors for real-time monitoring. The project is divided into two stages: hardware and software development. In the hardware section, two LDR sensors are used to detect the intensity of sunlight and determine the direction of maximum light. A servo motor is employed to adjust the position of the solar panel accordingly.

The system's performance has been tested and compared with that of a static solar panel. This project presents the design of a low-cost single-axis solar tracking system that enhances energy harvesting efficiency. The 555 IC serves as the main controlling unit. The LDR sensors detect the sun's position, and the servo motor adjusts the orientation of the solar photovoltaic (PV) panel.

All components, including sensors and the servo motor, are properly interfaced, and the motor is mechanically coupled with the PV panel. The complete system has been assembled and tested successfully. The solar tracker effectively adjusts the panel direction based on sunlight, thereby improving efficiency. The single-axis tracking system follows the sun's daily movement and ensures maximum utilization of solar energy.

CHAPTER 1

INTRODUCTION

The conversion of solar energy into electrical energy is one of the most promising and rapidly developing technologies in the field of renewable energy. It is clean, reliable, and requires very low maintenance, making it an environmentally friendly alternative to conventional energy sources. Solar energy has minimal ecological impact and is especially useful in remote and rural areas.

A photovoltaic (PV) panel is a device used to capture solar radiation and convert it into electrical energy. These panels consist of an array of solar cells, which are primarily made of silicon. When sunlight falls on these cells, the photovoltaic effect takes place, converting light energy into electrical energy.

It is well known that solar panels generate maximum energy when their surface is perpendicular to the sun's rays. However, stationary PV panels remain perfectly aligned with the sun only for a short duration during the day. To overcome this limitation and maximize energy output, solar tracking systems are used. These systems adjust the position of the solar panels according to the movement of the sun throughout the day.

There are several methods of solar tracking, including active, passive, chronological, and manual tracking systems. In an active tracking system, sensors are used to detect the position of the sun. These sensors control a motor that moves the solar panel so that it remains perpendicular to the sun's rays.

However, this system may not perform accurately during cloudy conditions, as sensors may fail to detect sunlight properly.

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Passive tracking systems operate based on temperature differences caused by sunlight. These systems use a low boiling point fluid that shifts due to heat imbalance, causing the panel to move. However, passive systems are less accurate and may respond slowly, especially during early morning conditions.

Chronological tracking systems are time-based systems in which the panel is moved at a fixed rate throughout the day. Since the sun moves at approximately 15 degrees per hour, the motor is programmed to rotate accordingly. This method is more accurate but consumes more power due to continuous motor operation.

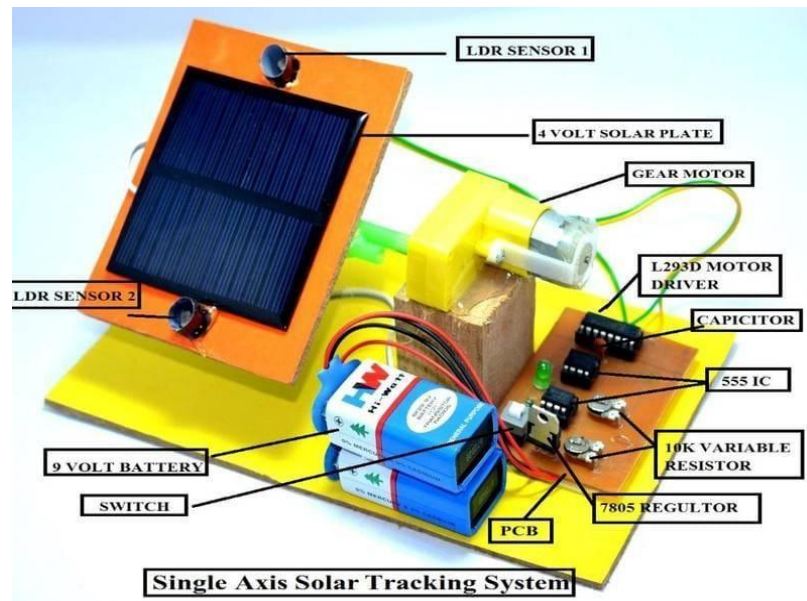
In manual tracking systems, human operators adjust the position of the solar panels periodically to follow the sun. Although simple, this method is less efficient and requires continuous effort.

Thus, solar tracking systems play a crucial role in improving the efficiency of solar panels by ensuring maximum exposure to sunlight throughout the day.

CHAPTER 2

SYSTEM ARCHITECTURE

Figure 1 shows the system architecture of the proposed single-axis solar tracking system.



The system consists of a photovoltaic (PV) panel mounted on a rotating shaft, which is driven by a bidirectional DC motor. The movement of the motor is controlled based on the real-time position of the sun, detected using two light intensity sensors (LDRs).

The two LDRs are placed on opposite sides of the solar panel and separated by an opaque barrier. Depending on the direction of sunlight, one LDR receives more light while the other remains partially shaded. The LDR exposed to higher light intensity produces a stronger electrical signal compared to the shaded one. This difference in output voltage between the two LDRs is used to determine the direction in which the panel should rotate.

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A 555 Timer IC is used as the main control unit to generate pulse signals for controlling the motor. Based on the signal difference from the LDRs, a relay circuit controls the direction of the motor,

Allowing it to rotate either clockwise or anticlockwise. This ensures that the solar panel continuously aligns itself perpendicular to the sun's rays for maximum energy absorption.

Solar energy is widely used across the world due to its availability, sustainability, and environmentally friendly nature. It does not require large-scale industrial setups and can be easily utilized in homes, agriculture, and industries panel. Since solar panels produce maximum power when sunlight falls directly on them, tracking systems are essential to improve efficiency.

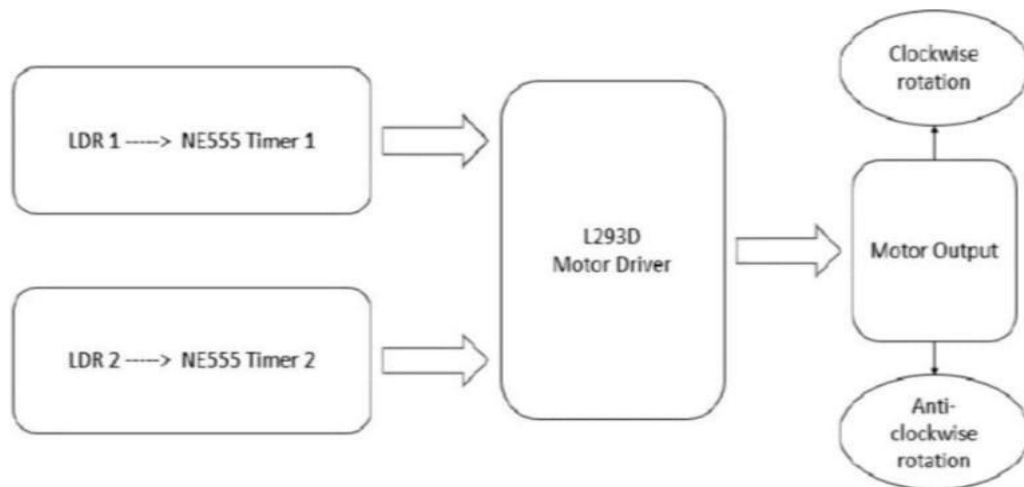
This project focuses on developing a single-axis solar tracker that automatically adjusts the panel orientation throughout the day. Studies show that such tracking systems can increase solar efficiency by approximately 30% to 40% compared to fixed panels.

Hardware Components Used:

1. D882 transistor
2. Light Dependent Resistor (LDR)-2
3. DC Motor (10 RPM)
4. DC gear motor
5. Solar Panel
6. 9V Battery
7. Switch
8. Resistors
9. Connecting Wires

CHAPTER 3

BLOCK DIAGRAM



The block diagram of the solar tracking system represents the main components and their functional relationship. The system uses a low-speed DC motor (10 RPM) to rotate the solar panel along a single axis.

Two Light Dependent Resistors (LDRs) are placed at both ends of the solar panel along the axis. These sensors detect the intensity of sunlight. Among the two LDRs, the one receiving higher light intensity generates a stronger signal. Based on this difference, the motor rotates the panel in the direction of higher light intensity until both LDRs receive equal light. This ensures maximum absorption of solar energy.

The single-axis solar tracking system consists of a photovoltaic (PV) panel mounted on a rotating shaft. The rotation is controlled by a bidirectional motor, which adjusts the panel position according to the sun's movement. The system continuously aligns the panel perpendicular to the sun's rays for improved efficiency.

Depending on the sunlight intensity, one LDR may be illuminated while the other is partially shaded. This creates a voltage difference between the two sensors. This voltage difference is used as a control signal to drive the motor, ensuring that the panel always faces the direction of maximum sunlight.

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WORKING OF SOLAR TRACKER

A solar tracking system is an automatic control system designed to follow the movement of the sun. It consists of components such as photovoltaic (PV) cells, sensors (LDRs), control units, motor drivers, and power supply systems.

When sunlight falls on the solar panel, the LDR sensors detect the light intensity. The sensor outputs are processed by the control circuit (using 555 Timer IC). Based on the difference in light intensity, the control unit generates signals to drive the motor.

The motor then rotates the solar panel in the direction where maximum sunlight is available. This process continues throughout the day, ensuring that the panel remains aligned with the sun. As a result, the solar tracker improves the efficiency of the solar panel by maximizing energy absorption compared to a fixed system.

CHAPTER 4

METHODOLOGY

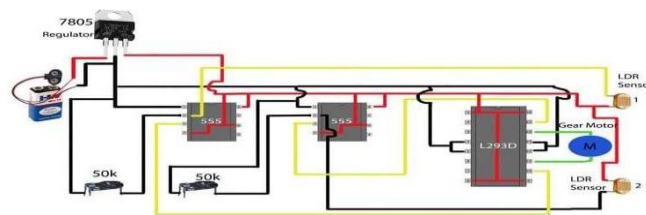
This work presents a photovoltaic (PV) solar tracker design methodology that improves the orientation and efficiency of the solar panel based on the geographical location (latitude) of installation.

The proposed system is experimentally validated by implementing a solar tracking system. The methodology focuses on developing a low-cost, high-availability tracking mechanism with acceptable accuracy. Compared to traditional systems, this design reduces cost and improves energy efficiency.

The system enhances solar energy collection by adjusting the panel orientation according to sunlight direction. Studies show that this type of system can increase efficiency significantly compared to fixed solar panels.

The methodology is based on the use of light sensors (LDRs), control circuits (555 Timer IC), and a motor mechanism to automatically align the solar panel with the sun throughout the day. This approach can also be extended to other solar tracking systems.

CIRCUIT DESIGN



The circuit consists of LDR sensors, a 555 Timer IC, L293D motor driver, and a DC motor. The LDR sensors detect sunlight intensity and generate signals based on light variation.

These signals are processed by the 555 Timer IC, which produces output pulses. The L293D motor driver receives these signals and controls the direction of the DC motor.

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Based on the difference in light intensity, the motor rotates the solar panel either clockwise or anticlockwise, ensuring maximum exposure to sunlight.

COMPONENTS OF SOLAR TRACKER

1. Solar Panel:



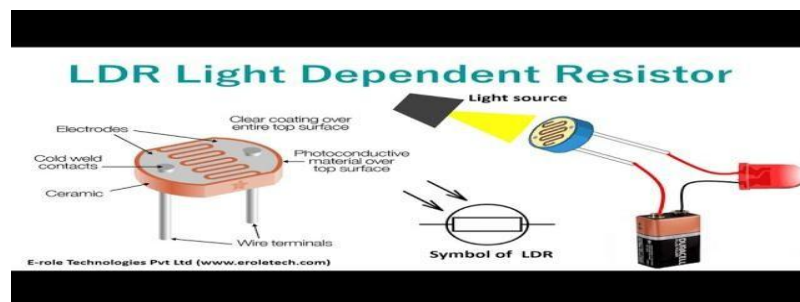
A solar panel is a device that converts sunlight into electrical energy. It is called a "solar" panel because it uses energy from the Sun, which is the most powerful and widely available source of light. Solar panels are also known as photovoltaic (PV) panels, where "photo" means light and "voltaic" means electricity.

A solar panel is made up of multiple solar cells arranged over a large surface area. These solar cells are usually made of silicon. When sunlight falls on the solar cells, it produces electrical energy through the photovoltaic effect.

The amount of electricity generated depends on the intensity of sunlight. The more light that falls on the panel, the more electricity is produced. For maximum efficiency, the surface of the solar panel should be kept clean and positioned to receive direct sunlight.

The basic structure of a solar cell includes a protective transparent coating on the top surface, electrodes (positive and negative), and a semiconductor material in between, which helps in the conversion of light into electricity.

2. LDR (Light Dependent Resistor):



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A Light Dependent Resistor (LDR), also known as a photoresistor or photoconductive cell, is a light-sensitive electronic component whose resistance changes according to the intensity of light falling on it.

The resistance of an LDR decreases as the intensity of light increases. In darkness, an LDR can have a very high resistance (in mega ohms), while in bright light, its resistance drops to a very low value (a few hundred ohms). This property is known as photoconductivity.

An LDR is made of a high-resistance semiconductor material such as Cadmium Sulfide (CdS).

When light falls on its surface, photons transfer energy to electrons, allowing them to move freely. This movement of electrons increases conductivity and reduces resistance.

In a solar tracking system, two LDR sensors are used to detect the direction of sunlight. These LDRs are placed on opposite sides of the solar panel. When one LDR receives more light than the other, it produces a lower resistance and a stronger signal. This difference in signals helps the system determine the direction of maximum sunlight.

Based on this difference, the control circuit activates the motor to rotate the solar panel toward the direction of higher light intensity. Thus, LDRs play a key role in improving the efficiency of the solar tracking system by ensuring maximum exposure to sunlight.

3.DC Gear Motor:



DC Gear Motor:- gear motor is an all-in-one combination of a motor and gearbox. The addition of a gear head to a motor reduces the speed while increasing the torque output. The important parameters

4. Capacitor:

A capacitor stores electrical energy temporarily and stabilizes voltage in the circuit.



A capacitor can store electric energy when it is connected to its charging circuit. And when it is disconnected from its charging circuit, it can dissipate that stored energy, so it can be used like a temporary battery. Capacitors are commonly used in electronic devices to maintain power supply while batteries are being changed.

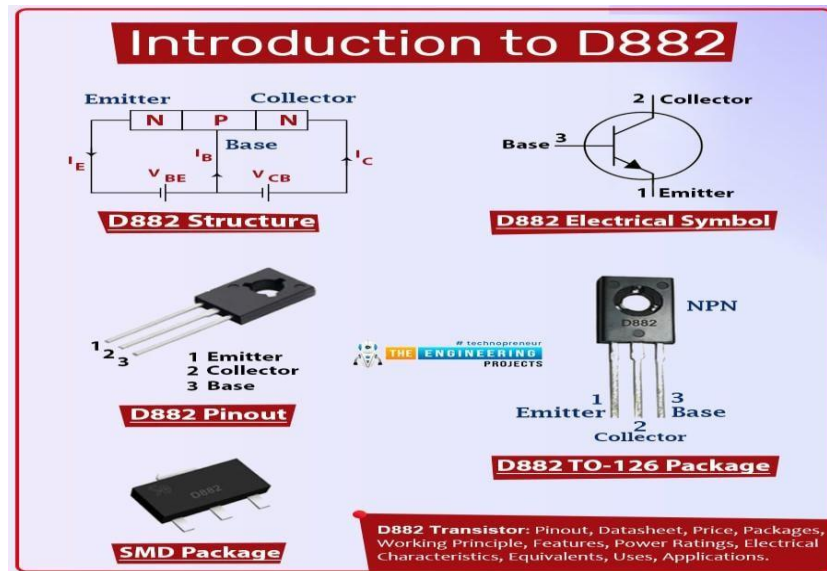
5. Resistor:



it is an electrical device may be a passive two-terminal electrical part that implements resistance as a circuit component. In electronic circuits, resistors unit of measurement accustomed reduce current flow, alter signal levels, to divide voltages, bias active components, and terminate transmission lines, among completely different uses.

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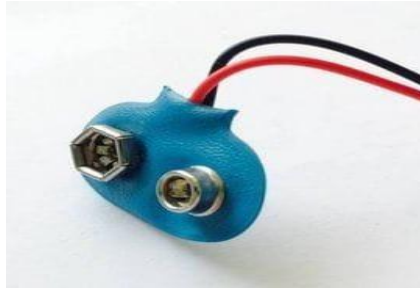
6.D882 Transistor



The D882 transistor, also known as 2SD882, is a medium power NPN bipolar junction transistor (BJT) commonly used in general purpose amplification and switching applications. It is designed with planer technology, offering reliable performance and moderate current handling capabilities. The transistor has three layers of semiconductor material with three terminals: emitter, base, and collector. It provides efficient current amplification with a gain range between 60 and 400, making it suitable for low power circuits. Additionally, the D882 can be mounted on a heatsink through the screw hole in its SOT-32 package, enhancing its heat dissipation during operation.

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7. Battery:



A battery assembled cap, a cylindrical battery with the cap and a method for making the same. The vent cap is attached to the battery cover by a hinge connection which allows for play between the vent cap and the battery cover and which allows for rotation of the vent cap.

8. Switch:

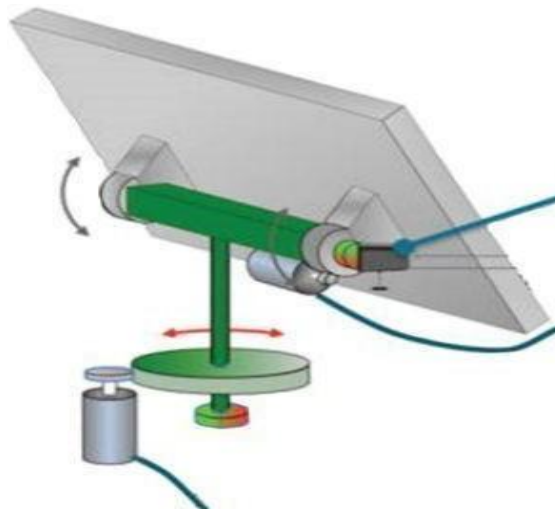


A switch, in the context of networking is a high-speed device that receives incoming data packets and redirects them to their destination on a local area network (LAN). A LAN switch operates at the data link layer (Layer 2) or the network layer of the OSI Model and, as such it can support all types of packet protocols.

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CONSTRUCTION

Two CDS cells are connected to port RB2 and RB3 of the PIC. the tw photocells are positioned on a small straight piece of wood or plastic. Another piece is mounted perpendicular to the straight piece, thereby dividing both the sensors. If both the photocells are equally illuminated by the sun, their resistance level will be same. As long as the resistance is same, an error margin of +10 points, the PIC will analyze this data and thus will not generate any signal to actuate the motor. if one of the sensor comes under a shadow, then the PIC will detect this change and thus it will actuate the motor to move the sensor module to a position where equal light is being illuminated on both of them. The PIC is programmed so that it can obtain its resistance data from the two LDRs and to move motor either clock wise or anti clock wise depending on which LDR is under shadow. The concept of the software design is dependent on the LDR that is under shadow.



OPERATION OF SOLAR TRACKER

The solar tracker operates based on the difference in light intensity detected by two Light Dependent Resistors (LDRs).

When LDR2 receives more light than LDR1, its resistance decreases and becomes lower than that of LDR1. This creates a voltage difference, which is given as input to the comparator circuit. As a result, the output signal activates the motor to rotate in one direction (anticlockwise), moving the solar panel toward the direction of higher light intensity.

Similarly, when LDR1 receives more light than LDR2, its resistance becomes lower. This reverses the voltage difference, causing the comparator to generate an output that rotates the motor in the opposite direction (clockwise). Thus, the solar panel continuously adjusts its position toward maximum sunlight.

The system uses a 555 Timer IC as part of the control circuit. It processes the signals received from the LDRs using an RC (resistor-capacitor) timing mechanism. Based on the comparison of inputs, the circuit generates output signals that control the motor driver.

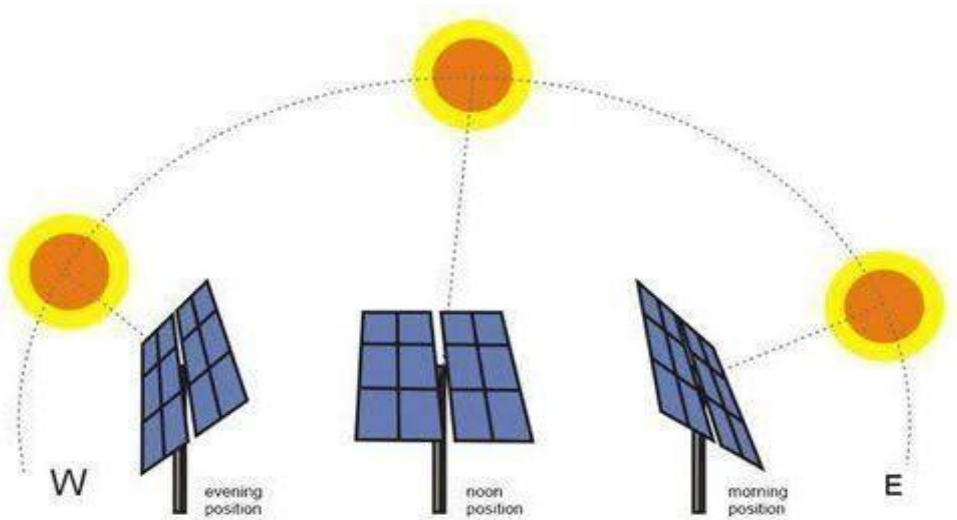
The motor driver (L293D) receives these signals and drives the DC motor in the required direction. Logic signals (HIGH and LOW) control the switching of transistors, allowing current to flow through the motor and enabling its rotation.

The solar tracker structure is built using materials such as PVC pipes, a wooden base (MDF), gears, and a DC motor. The motor is connected to a gear mechanism that rotates the panel smoothly. The system is designed to provide high torque at low speed, ensuring efficient and stable movement.

As the sun moves throughout the day, the LDR sensors continuously detect light intensity and adjust the panel position accordingly. This ensures that the solar

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panel remains perpendicular to the sun's rays, thereby maximizing energy efficiency.



A solar tracker positions the solar panels at an angle directed to the sun. It is an advanced sun monitoring system that can rotate the panels to track the movement of the sun across the sky.

It facilitates the panel system to trap the maximum sunlight and optimise the energy output. There are considerable advantages to using a solar energy tracker.

CHAPTER 5

ADVANTAGES OF TRANSISTOR SOLAR TRACKER

- Single-axis trackers move from east to west, following the sun's direction.
- They are simple in design and easy to install.
- Lower cost compared to dual-axis tracking systems.
- Require less maintenance than complex tracking systems.
- More reliable due to fewer moving parts.
- Have a longer lifespan compared to dual-axis trackers.
- Suitable for users looking for a cost-effective solution.
- Provide higher efficiency compared to fixed solar panels (around 30-35% improvement).
- Can work effectively even in areas with moderate sunlight.

DISADVANTAGES OF TRANSISTOR SOLAR TRACKER

- More expensive than fixed (stationary) solar panel systems.
- Requires periodic maintenance due to moving parts.
- Installation cost is higher than fixed systems.
- Less efficient compared to dual-axis trackers.
- Performance may reduce during cloudy weather.

APPLICATIONS OF TRANSISTOR SOLAR TRACKER

- Used in agriculture fields for water pumping systems.
- Used in domestic applications for lighting purposes.
- Used in solar power plants for electricity generation.
- Used in street lighting systems.
- Used in industries requiring renewable energy solutions.

CHAPTER 6

CONCLUSION

This project successfully demonstrates the design and implementation of a solar tracking system that improves the efficiency of solar energy generation. It is well established that a single-axis solar tracking system can increase energy output by approximately 20%-30%, while dual-axis systems can further enhance efficiency.

In this project, a solar tracking system is developed using LDR sensors, a 555 Timer IC, and a motor control mechanism. The system automatically detects the direction of maximum sunlight and adjusts the position of the solar panel accordingly. This ensures that the panel remains perpendicular to the sun's rays for maximum energy absorption.

The performance of the system has been tested and compared with a fixed solar panel, and the results show improved efficiency and better utilization of solar energy. The use of simple and low-cost components makes this system affordable and practical for real-world applications.

Although some challenges such as weather conditions and system calibration were encountered during testing, the overall implementation was successful. The system provides stable and precise movement of the solar panel.

In conclusion, the solar tracking system is an effective solution for enhancing solar energy generation and can be widely used in domestic, agricultural, and industrial applications.

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